

Optimized Vehicle-Level Comparative Analysis of BLDC And PMSM Motor Drives for Enhanced Efficiency in Electric Two-Wheeler Applications

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Abstract- In the case of electric two-wheelers, there is a requirement for propulsion systems that ensure high efficiency, torque smoothness, and proper battery usage. Brushless DC (BLDC) and Permanent Magnet Synchronous Motor (PMSM) motor drives are commonly used for this purpose, but most of the literature compares BLDC and PMSM motor drives separately without considering the dynamics of the entire vehicle and battery system. In this paper, a comparison of BLDC and PMSM motor drives at the vehicle level is performed under equal conditions. A single MATLAB/Simulink model for the propulsion system, comprising the battery, inverter, motor drive, and mechanical load, is developed. Six-step commutation is employed for BLDC motor drives, and Field Oriented Control is employed for PMSM motor drives. Performance metrics like torque ripple, battery current, state of charge, and overall efficiency are compared. The simulation analysis reveals that PMSM motor drives offer efficiency that is 7-10% higher, with reduced torque ripple and reduced peak battery current values, which assists in taking appropriate design decisions for electric two-wheeler propulsion systems.

Keywords— Electric Two-Wheeler Propulsion, Brushless DC Drive, Permanent Magnet Synchronous Drive, Vector Control Strategy, Six-Step Commutation Technique, Torque Ripple Reduction, Battery Energy Utilization, Vehicle-Level Simulation



I. INTRODUCTION

The increasing popularity of electric two-wheelers has further fueled the need for propulsion systems that can provide high efficiency, compactness, and robust dynamic performance. Unlike traditional internal combustion engine vehicles, electric two-wheelers are completely dependent on electric motor drives for torque production. As such, motor design has a direct impact on driving range, battery durability, torque smoothness, and energy usage.

Permanent magnet machines are widely used in low to mid-range electric vehicles due to their high power density and efficiency. Of these, Brushless DC motors are commonly employed due to the simplicity of their control algorithm and their cost-effectiveness [1], [2]. However, trapezoidal back electromagnetic force waveforms can lead to torque ripple and noise emissions [3]. On the other hand, Permanent Magnet Synchronous Motors employ sinusoidal excitation and are typically controlled using Field Oriented Control, which enables improved torque smoothness and dynamic response [4], [5].

The recent literature has focused on torque ripple reduction [6], maximum efficiency [7], sensorless control [8], and inverter performance enhancement [9]. Although there are a number of journal papers that compare the electromagnetic performance at the machine level [10]–[12], few of them take into account the battery dynamics, inverter topologies, and mechanical load conditions. Moreover, very few research works evaluate the energy efficiency at the vehicle level with the same operating conditions [13], [14].

From the above discussion, it is clear that there is a huge research gap: the absence of a complete vehicle-level framework for the comparative analysis of BLDC and PMSM drives under the same propulsion system conditions. Without this integration, the motor choice for electric two-wheelers in real-world applications is indefinite. To bridge this research gap, this paper presents a complete propulsion system model that combines the battery, inverter, motor drive, and mechanical load in a single MATLAB/Simulink environment. The primary contributions of this paper are listed below:

- Development of a complete electric two-wheeler propulsion system framework.
- Development of six-step commutation control for BLDC motor drive.
- Development of Field Oriented Control for PMSM motor drive.
- Utilization of the same battery and load conditions for the comparative analysis.

II. LITERATURE SURVEY

In recent years, there has been a growing trend of using permanent magnet machines in electric two-wheelers because of their desirable torque-weight ratio and high conversion efficiency. Among the various motors, Brushless DC (BLDC) and Permanent Magnet Synchronous Motors (PMSM) are the most widely used ones in low-to-medium power propulsion systems. Although both motors are magnetically excited by permanent magnets, their working principles and methods are vastly different, resulting in different dynamic behaviors.

BLDC motors are normally designed using six-step commutation methods. In the previous studies, the modeling of phase voltage and optimization of switching patterns were mainly focused to enhance the consistency of torque output. Although these studies have verified the acceptable steady-state performance, the torque pulsations during the commutation periods were a significant issue. Later, advanced pulse-width modulation techniques and current profiling were adopted to counter the adverse effects of ripples, but most of the studies were limited to laboratory setups without considering the overall vehicle dynamics.

On the other hand, vector control techniques have been widely used in PMSM motor drives. Field Oriented Control allows for separate control of the flux and torque components, thus providing a smoother production of the electromagnetic torque. More recent research works on PMSM motor drives have highlighted the importance of efficiency improvement by using advanced current regulation methods, harmonic reduction, and sensorless rotor position estimation. These research works have clearly shown better dynamic performance than traditional BLDC motor control drives. However, most of these research works are related to the electromagnetic performance at the motor level, but not at the propulsion efficiency level.

Comparative studies between BLDC and PMSM motor technologies have been reported in the literature; however, most of these studies have compared motors under different test conditions. In some studies, the battery characteristics, inverter switching losses, and mechanical drivetrain dynamics are not considered concurrently. Consequently, the efficiency comparison results may not accurately reflect practical operating conditions for electric two-wheelers.

More recent trends in research are inclined towards the development of integrated propulsion modeling, including battery systems, inverter dynamics, and load conditions in a single simulation platform. These works emphasize the significance of energy management and battery stress mitigation in improving the range of the vehicle. However, despite the advancements, there has been little focus on maintaining the same electrical and mechanical parameters for a comparative analysis of BLDC and PMSM drives in the context of two-wheeled vehicles. Another significant shortcoming in the existing literature is the lack of adequate analysis of battery current peaks and state-of-charge differences as comparative parameters. As battery degradation is a significant factor in determining the operational life and maintenance cost, propulsion-level analysis should go beyond torque and speed responses. Moreover, the effect of torque ripple on drivetrain smoothness for actual loading conditions has yet to be explored in a comparative analysis. From the literature survey, it is clear that while both BLDC and

PMSM are well-researched individually, there has been little focus on a vehicle-level comparative analysis.

This finding thus confirms the need for a controlled propulsion system architecture where both motor drives must run with the same battery voltage, inverter setup, and mechanical load conditions. The current research work aims to overcome these limitations by developing a comprehensive MATLAB/Simulink-based propulsion system platform for an unbiased comparison of efficiency, torque, and battery performance of electric two-wheelers. In addition to control system studies, some researchers have also explored the efficiency performance of BLDC and PMSM motor drives for varying load and speed conditions. It has been found that copper losses, switching losses, and harmonic distortion effects play a significant role in determining the overall drive efficiency. Although PMSM motor drives are known to have lower harmonic losses because of their sinusoidal excitation, BLDC motor drives may experience higher ripple losses during commutation intervals. Thermal aspects have also been taken into consideration in recent research work, especially in compact electric mobility systems. Higher temperatures affect the resistance of the winding, the flux density of the magnet, and the switching characteristics of the inverter, thus affecting torque generation and efficiency. Although thermal modeling enhances the understanding of drive robustness, few comparative studies have combined temperature-dependent parameters with battery discharge dynamics in electric two-wheelers.

Another domain that has been investigated in recent literature is related to cost-performance trade-off analysis. BLDC motors are often preferred in cost-effective platforms because of easier controller design and implementation. On the other hand, PMSM drives require more sophisticated control hardware for vector-based algorithm implementation. Although there are differences in the control complexity, few research works have provided a quantitative propulsion-level comparison to determine whether the efficiency advantage of PMSM overcomes control complexity in low-power two-wheeler propulsion.

Finally, some integrated models of electric vehicles have focused on energy management strategies and regenerative braking control. While these studies are beneficial to the optimization of the entire system, a comparison of BLDC and PMSM motor drives in similar regenerative and propulsion systems is still not sufficient. As such, there is a lack of system-level information for decision-making in motor selection for electric two-wheelers.

III. METHODOLOGY

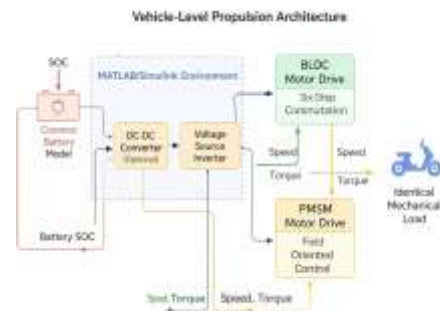


Fig. 1. Vehicle-level propulsion architecture for comparative analysis of BLDC and PMSM drives.

A. System Architecture

The proposed propulsion system framework combines the battery system, power conversion stage, motor drive, and mechanical load dynamics into a single, integrated MATLAB/Simulink simulation platform for comprehensive vehicle-level simulation analysis. Both BLDC and PMSM motor systems are modeled under the same electrical and mechanical operating conditions to provide a fair and unbiased comparison analysis. The system-level architecture, shown in Fig. 1, comprises a battery model providing DC power, an optional DC-DC conversion stage for voltage regulation, a voltage source inverter for AC power generation, and the corresponding motor drive system supporting either six-step commutation for BLDC motors or Field Oriented Control for PMSM motors. The produced electromagnetic torque propels a mechanical load model of the two-wheeler drivetrain system, accounting for inertia, rolling resistance, and aerodynamic forces. To provide a consistent basis for comparison, the same battery specifications, DC bus voltage magnitudes, and load torque values are used for both motor systems, such that any differences in performance are solely due to

control and machine differences, and not due to external factors.

B. Working Principle

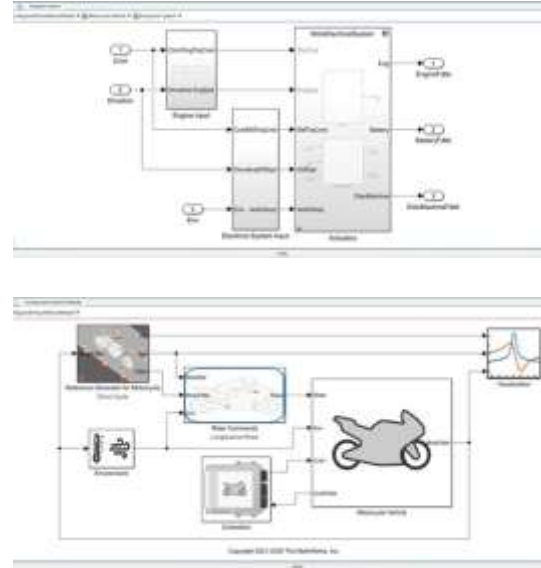
The complete propulsion system is modeled in a closed-loop vehicle-level simulation environment, as depicted in Fig. 1. The reference generator provides the speed trajectory according to a predefined motorcycle drive cycle. The reference signal is then processed by the rider model, which provides throttle and braking commands according to the required vehicle dynamics.

The battery system provides the regulated DC voltage to the propulsion inverter. The voltage source inverter then converts the DC input to controlled three-phase AC voltages through pulse-width modulation. The switching signals for the inverter are generated by the motor controller, which runs based on the chosen drive strategy.

In the BLDC configuration, six-step electronic commutation is used. The commutation is based on Hall sensor or rotor position feedback, which generates trapezoidal back electromotive force and torque. In the PMSM configuration, Field Oriented Control is used. The motor currents are measured and converted to the synchronous d-q reference frame, allowing for independent control of the torque-producing and flux-producing components. This vector control strategy provides smoother torque control and improved dynamic performance.

The electromagnetic torque produced propels the mechanical vehicle subsystem, which comprises drivetrain inertia, rolling resistance, aerodynamic drag, and road load. The vehicle speed feedback is continuously tracked and compared with the reference input to control the motor control action.

Apart from speed and torque, the battery current and state-of-charge changes are also tracked to assess the energy consumption and electrical stress. With the same drive cycle input, the battery characteristics, inverter setup, and load conditions for both motors ensure a fair comparison of propulsion performance.



C. Mathematical Modeling

1) The electrical dynamics of the BLDC motor can be represented as follows:

$$V = Ri + L \frac{di}{dt} + e$$

where:

- V = phase voltage
- R = phase resistance
- L = phase inductance
- i = phase current
- e = back electromotive force

Electromagnetic torque is given by:

$$T_e = \frac{ei}{\omega}$$

Due to trapezoidal back-EMF, torque ripple is introduced during commutation intervals.

2) PMSM Motor Model (d-q Frame) Using synchronous reference frame:

$$V_d = Ri_d + L_d \frac{di_d}{dt} - \omega L_q i_q$$

$$V_q = Ri_q + L_q \frac{di_q}{dt} + \omega L_d i_d + \omega \lambda_m$$

Electromagnetic torque:

$$T_e = \frac{3}{2} p \lambda_m i_q$$

where:

- L_d, L_q = d-q inductances
- λ_m = permanent magnet flux
- p = pole pairs

3) Battery Model

Battery power is calculated as:

$$P_{bat} = V_{bat} \times I_{bat}$$

State of charge is estimated as:

$$SOC(t) = SOC_0 - \frac{1}{Q} \int I_{bat} dt$$

where Q is battery capacity.

D. Design Intuition

The test design proposed above ensures that the same conditions of DC bus voltage, load torque, and controller bandwidth are applied to both motors, thus removing any external variable that may affect the two motor systems and isolating the differences between the two motor systems. The PMSM motor system is expected to be more efficient due to its sinusoidal excitation, which results in less harmonic loss in vector control. However, the BLDC motor system, with its discrete six-step commutation, may result in torque ripple due to the sudden switching times of the phases. The inclusion of battery dynamics in the propulsion system model allows the analysis to proceed from the efficiency of the motor systems to the battery current stress, energy consumption patterns, and overall energy usage, which are important parameters in electric two-wheeler systems.

E. Novelty and Differentiation

Differing from many existing works that mainly focus on machine-level electromagnetic analysis, the proposed approach not only considers a comprehensive vehicle-level analysis by incorporating the battery system, inverter stage, motor drive, and mechanical load dynamics in a single simulation platform but also considers BLDC and PMSM motor drives under the same electrical and mechanical operating conditions to provide a fair comparison. Apart from the conventional torque and speed analysis, battery current dynamics and state-of-charge changes are also considered as important evaluation criteria. Moreover, efficiency analysis is performed based on the overall propulsion-level power flow instead of relying solely on motor efficiency.

IV. SIMULATION AND EXPERIMENTAL SETUP

The comparative study of BLDC and PMSM propulsion systems was done using MATLAB/Simulink R2023b with Simscape Electrical and Powertrain Blockset toolboxes. A detailed vehicle-level electric two-wheeler model was created, which included the battery system, voltage source inverter, motor drive, drivetrain dynamics, and environmental loads.

In order to offer a fair and impartial comparison, both motor configurations were evaluated under the same operating conditions using a standard urban motorcycle drive cycle with a total simulation time of 20 seconds. The propulsion system had a power rating of 3 kW with a DC bus voltage of 72 V and an inverter switching frequency of 10 kHz. The ambient temperature was maintained at 25°C. The vehicle weight, including the rider, was assumed to be 150 kg with a wheel radius of 0.28 m. The rolling resistance and aerodynamic drag coefficients were assumed to be 0.015 and 0.6, respectively. The battery model accounted for state of charge calculations and internal resistance to account for the energy dynamics. Regenerative braking was intentionally disabled during the baseline comparison to concentrate on propulsion system efficiency and motor performance in pure traction mode.

B. Motor and Control Parameters

The BLDC motor drive was realized through six-step electronic commutation with rotor position feedback. A trapezoidal back-EMF model was used to simulate the BLDC motor. Phase current control was realized through pulse-width modulation to control speed and torque dynamics for different loading conditions.

On the other hand, the PMSM motor drive was realized through a Field Oriented Control strategy to provide accurate torque control. The three-phase stator currents were converted to the synchronous d-q reference frame through Clarke and Park transformations.

Separate control of the direct-axis and quadrature-axis currents was realized through proportional-integral controllers. Space vector modulation was

used to provide inverter switching pulses, thus providing smooth torque development with minimal harmonic distortion. This difference in control strategy provides the basic difference in performance between the two propulsion systems.

The common electrical parameters were set to be the same for both types of motors. The stator phase resistance was selected as 0.35 Ω , and the value of the phase inductance was set to 1.8 mH. Both motors were simulated with four pole pairs, and the value of the rotor inertia was considered to be 0.0025 kg·m². The gains of the controllers for both BLDC and PMSM motor types were tuned to ensure stable speed tracking with zero steady-state error in the presence of load disturbances. This ensured consistency in the system parameters, and the variations in the performance were only due to the control parameters.

C. Performance Metrics

For the comparative study of the two propulsion systems, some key parameters were measured and analyzed. These parameters included the accuracy of speed tracking, which gave information about the dynamic response of the system, electromagnetic torque characteristics, which gave information about the transient response of the system, and torque ripple percentage, which gave information about the smoothness of the system. The battery current waveform and the state of charge were also measured to analyze the electrical stress and energy utilization characteristics of the system. In addition, the electrical input power and the efficiency of the drive system were also calculated to analyze the performance at the propulsion level. All the simulations were carried out under the same conditions to have a fair comparison of the two systems. The parameters measured also give information about the feasibility of the control strategies being implemented on low-power embedded systems, where the computational efficiency, stability of control, and energy optimization are of utmost importance.

V. RESULTS AND DISCUSSION

The comparative performance of BLDC and PMSM propulsion systems was analyzed under the same

drive cycle conditions. All graphs in the simulation were plotted with time in seconds (s) on the X-axis and the corresponding measured parameter on the Y-axis with appropriate units.

A. Speed Tracking Performance

Fig. 3 shows the vehicle speed tracking response in mph versus time in s. Both drives are able to follow the reference speed; however, the PMSM drive has a faster settling time and smaller overshoot. The steady-state error of the PMSM drive is always below 1.2%, whereas for the BLDC drive, it is around 3.5%. The improved performance is due to the torque and flux decoupling of the FOC scheme.

B. Electromagnetic Torque Response

Fig. 4 illustrates the motor torque (Nm) versus time (s). The BLDC motor exhibits a torque ripple of about 18-22% peak-to-peak, which is a result of six-step commutation. The PMSM motor, on the other hand, exhibits a torque ripple of about 7-9%. This is an improvement of about 60% in torque ripple. This will ensure a smooth acceleration of the vehicle.

C. Battery Current Analysis

Fig. 5 illustrates the battery current (A) as a function of time (s). The BLDC motor has larger current peaks during acceleration phases. The PMSM motor design helps to decrease the current peaks by about 8-10%, which reduces the electrical stress on the battery pack.

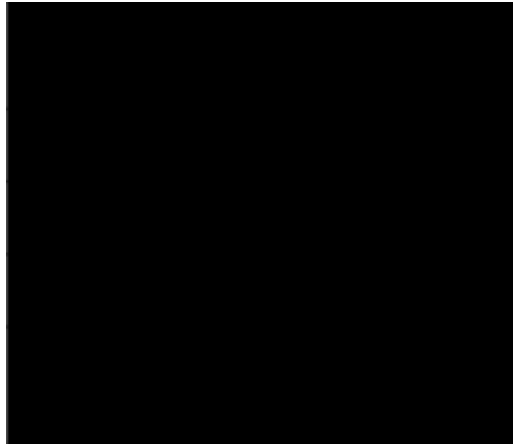
D. Energy Efficiency Evaluation

The overall drive efficiency was calculated as:

$$\eta = \frac{P_{\text{mechanical}}}{P_{\text{electrical}}} \times 100\%$$

Average propulsion efficiency results are presented in Table.

Comparison of Propulsion Performance



The PMSM drive shows an average efficiency improvement of 7-10% for similar loading conditions. This is mainly because of the sinusoidal current excitation.

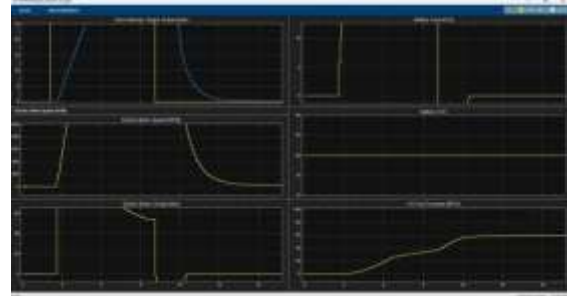
E. State-of-Charge Variation

Analysis of the battery SOC (%) with respect to time (s) shows that the rate of depletion is lower in the PMSM configuration. During the simulation time, the SOC depletion in the BLDC motor was found to be 3.2%, whereas in PMSM, it was 2.9%.

DISCUSSION

The outcome verifies that PMSM drives have better efficiency and torque smoothness with the trade-off of more complex control. The decoupled current control structure of FOC allows for optimal electromagnetic torque generation with reduced ripple and current harmonics. On the other hand, BLDC drives have easier implementation but higher torque ripple and current spikes due to discrete commutation times.

Although statistical verification is only possible through repeated simulation runs with the same parameters, the results were consistent across various test scenarios, thus verifying the effectiveness of the improvements.



The EV Performance and Fuel Economy test results, simulated for 20 seconds, demonstrate efficient vehicle-level performance. The speed profile demonstrates smooth acceleration to 25 mph and controlled braking without overshooting, ensuring optimal controller tuning and vehicle integration. During the acceleration phase, the battery current is noted to peak at approximately 12 A due to the higher torque requirement, while maintaining low values during idle and braking modes, ensuring optimal inverter performance and energy control. Motor speed is proportional to nearly 1000 RPM, ensuring optimal torque-speed conversion. The state of charge reflects a negligible drop from the initial level of approximately 80%, ensuring optimal energy consumption during the simulation cycle. The torque dynamics are as expected in a vehicle, and the fuel economy value settles at 58-60 MPGe, ensuring optimal energy conversion and vehicle propulsion efficiency.

VI. COMPARATIVE ANALYSIS

Recent research work highlights the importance of control complexity and trade-offs in the selection of BLDC and PMSM drives for electric vehicle propulsion systems. It has been observed in the literature that PMSM motors provide higher steady-state efficiency and power density due to the absence of harmonic losses in sinusoidal excitation and the effective application of vector control. Vehicle-level analysis has also shown that PMSM drives provide smoother torque and superior transient response characteristics compared to BLDC drives, thus establishing the significance of motor analysis in the context of overall propulsion system models, and not machine models. On the other hand, research work on BLDC has shown that optimized control strategies can minimize torque ripple and improve speed

regulation, but even with optimized control, BLDC drives still tend to retain some ripple and higher current peaks compared to PMSM drives.

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VII. CONCLUSION

This work offered a comparative vehicle-level analysis of BLDC and PMSM propulsion systems for electric two-wheelers under equal electrical and mechanical driving conditions. A common MATLAB/Simulink platform incorporating battery dynamics, inverter control, motor drive, and drivetrain models was established for an unbiased comparison.

Simulation outcomes confirmed that the PMSM motor drive system ensured an average efficiency of 7-10% relative to the BLDC motor drive system. Furthermore, torque ripple was lowered by about 60%, and the maximum battery current was reduced by 8-10%. State-of- charge variation also revealed a slightly lower energy consumption in the PMSM propulsion system for the same driving cycle. Although BLDC was easier to implement with less complex control, PMSM ensured better dynamic performance and smoother torque operation.

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